A Reinforcement Learning Model for Anticipatory Shivering

Foundations of Computational Neuroscience - Project 4

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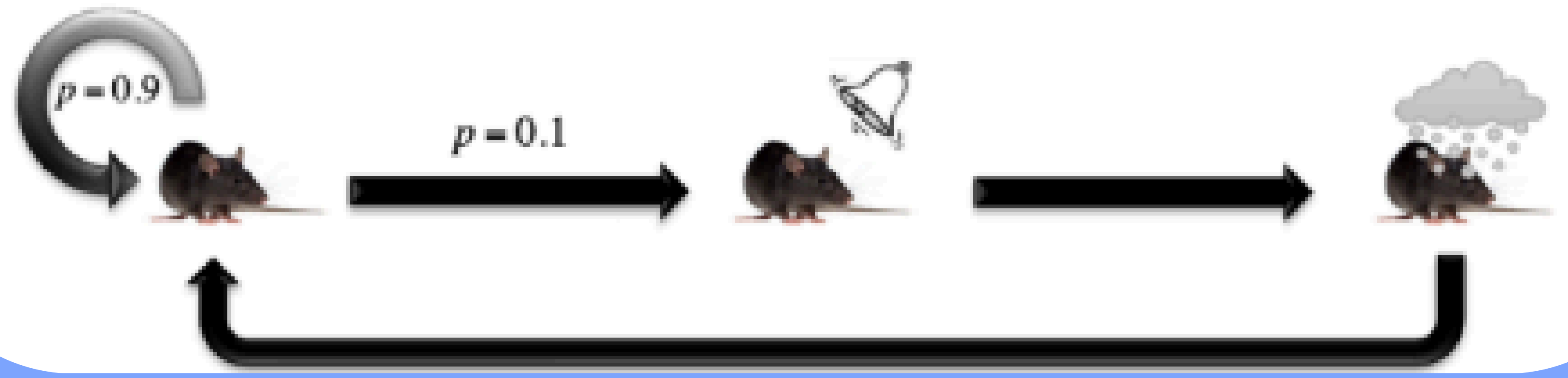
Our Framework

Classical conditioning experiment with a rat:

initial state

bell state

cold shower state



Before conditioning:

- rat shivers only when in the cold shower state
- instinctive response to cold

After conditioning:

- the rat starts shivering at the bell state, just after hearing the bell (anticipatory shivering)

Task

Describe and code an RL model that explains behaviors such as this anticipatory response

Literature References

- [1] Mehdi Keramati and Boris S. Gutkin. A reinforcement learning theory for homeostatic regulation. In Advances in Neural Information Processing Systems 24 (NeurIPS 2011), pages 82–90. Curran Associates, Inc., 2011
- [2] N. Keramati and B. Gutkin. Homeostatic reinforcement learning for integrating reward collection and physiological stability. PLoS Computational Biology, 10(11):e1003763, 2014
- [3] Naoto Yoshida, Henning Sprekeler, and Boris Gutkin. Linking homeostasis to reinforcement earning: internal state control of motivated behavior. Current Opinion in Behavioral Sciences, 66:101611, 2025

Current Opinion in Behavioral Sciences

Volume 66, December 2025, 101611

Review

Linking homeostasis to reinforcement learning: internal state control of motivated behavior

Naoto Yoshida [1], Henning Sprekeler [2 3 4 5], Boris Gutkin [6]

Homeostatically Regulated Reinforcement Learning (HRRL)

Reward as Drive Reduction

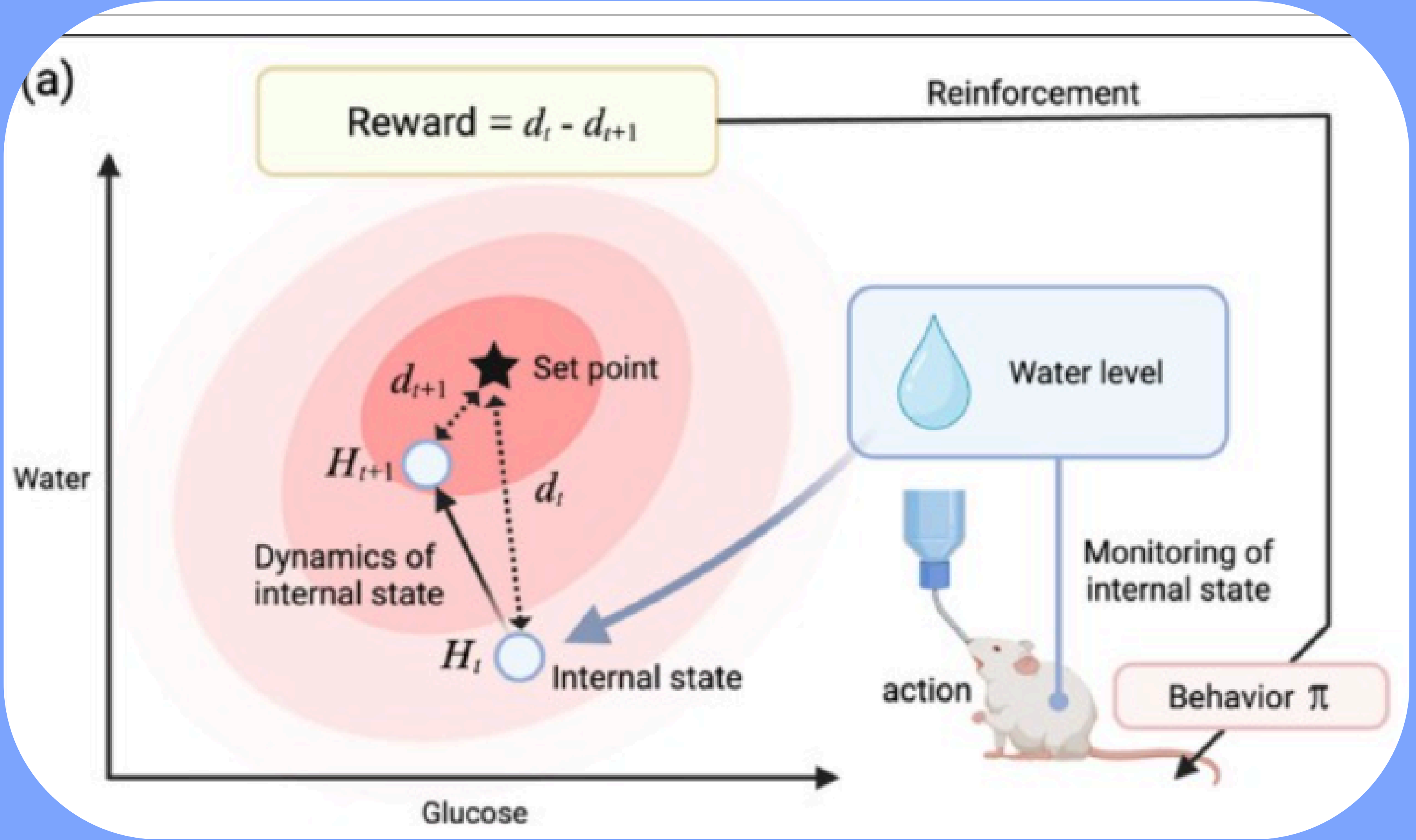
$$d(t) = (T(t) - T^*)^2$$

$$R_t = d(t) - d(t+1)$$

Effect of Shivering and Cold Shower on Rat Temperature

$$T(t+1) = T(t) + T_{\text{shiver}}$$

$$T(t+1) = T(t) - T_{\text{cold}}$$



? How to deal with:

- Actions being independent of next state
- Encoding the internal temperature:
  - as not part of the state-action space
  - affecting the reward
- The ‘natural’ decay towards the optimal temperature (independent of shivering)

Challenges

two-step SARSA algorithm

$$R_{t+1} + \gamma R_{t+2} + \gamma^2 Q(S_{t+2}, A_{t+2})$$

Q-value update

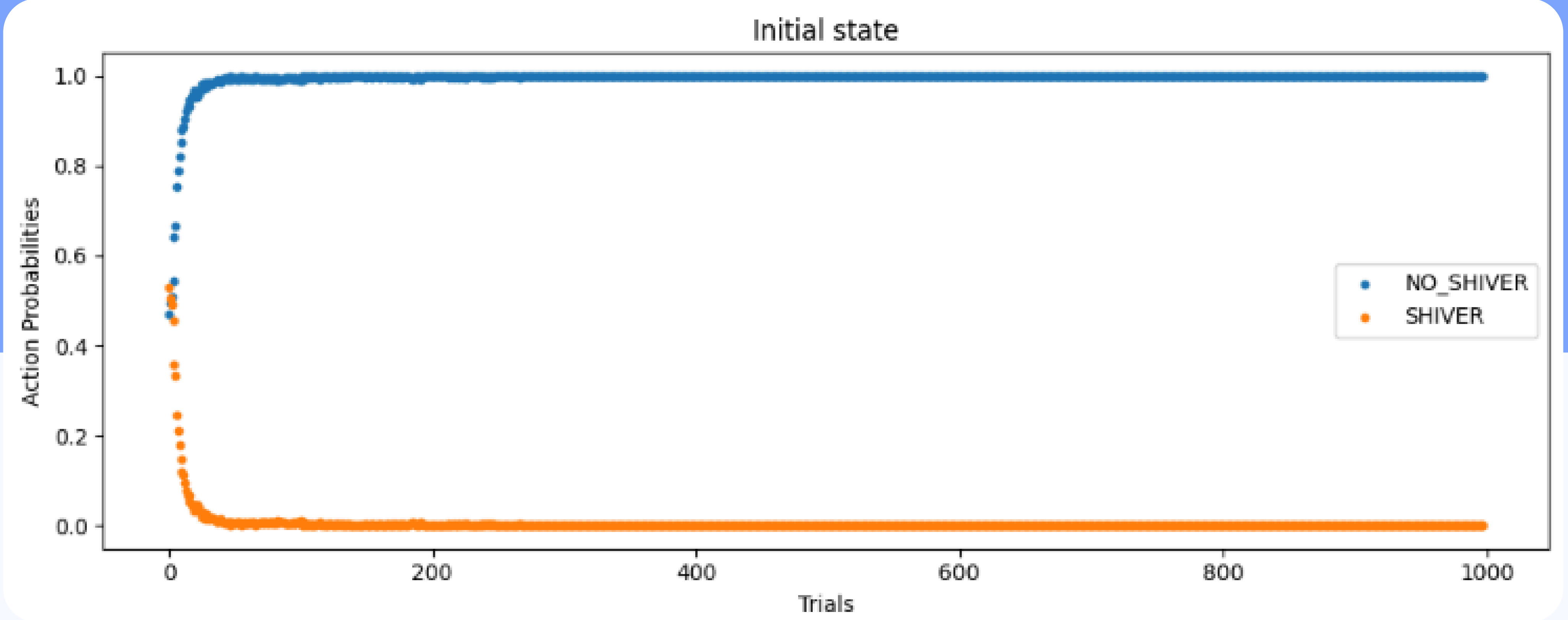
$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \left(R_{t+1} + \gamma R_{t+2} + \gamma^2 Q(S_{t+2}, A_{t+2}) - Q(S_t, A_t)\right)$$

Softmax action selection
(SHIVER vs NO_SHIVER)

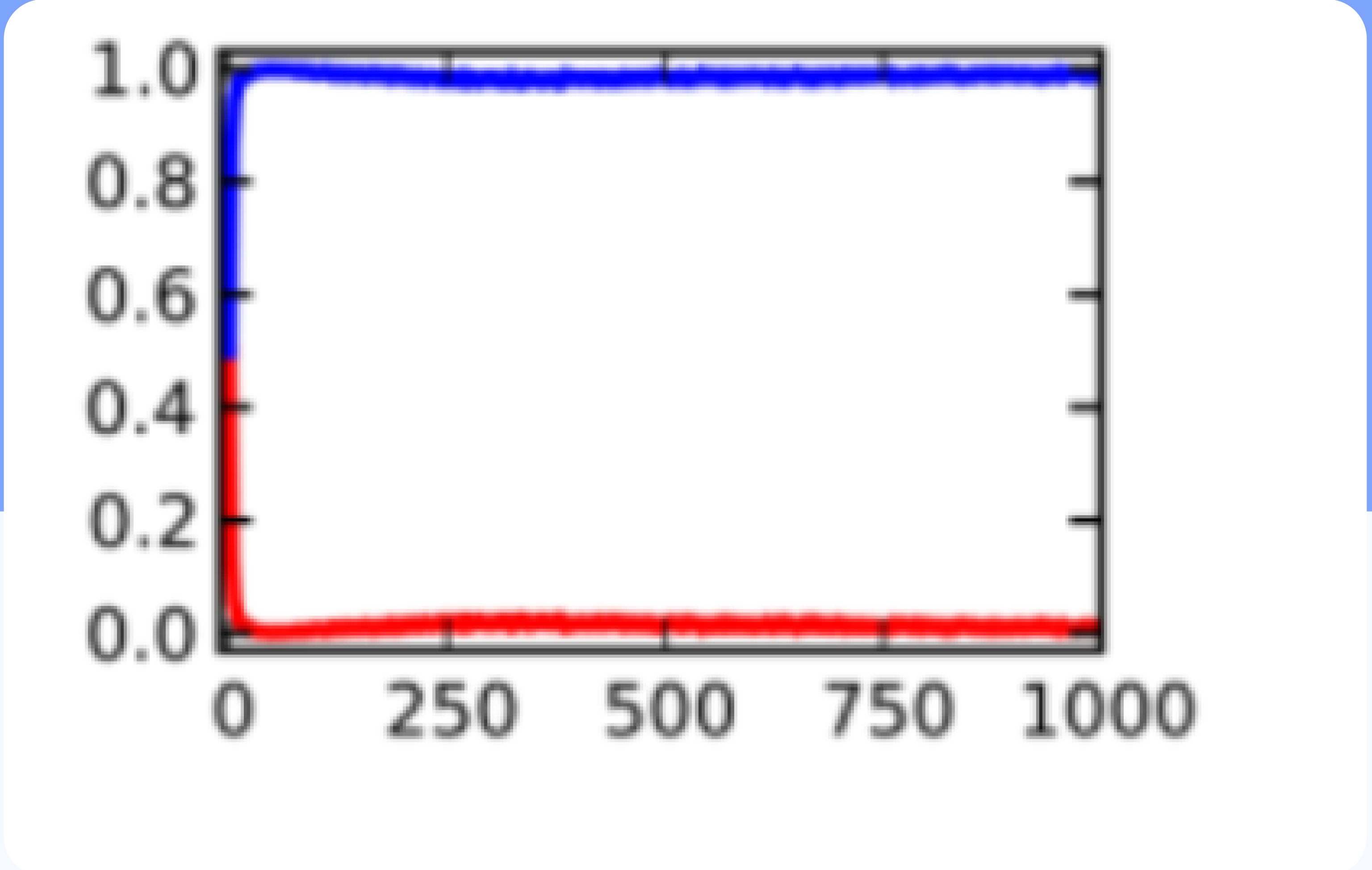
$$\pi(a_i \mid s) = \frac{e^{Q(s,a_i)/\tau}}{\sum_{j=1}^{n} e^{Q(s,a_j)/\tau}}$$

RL Coding Implementation

Results - Initial State

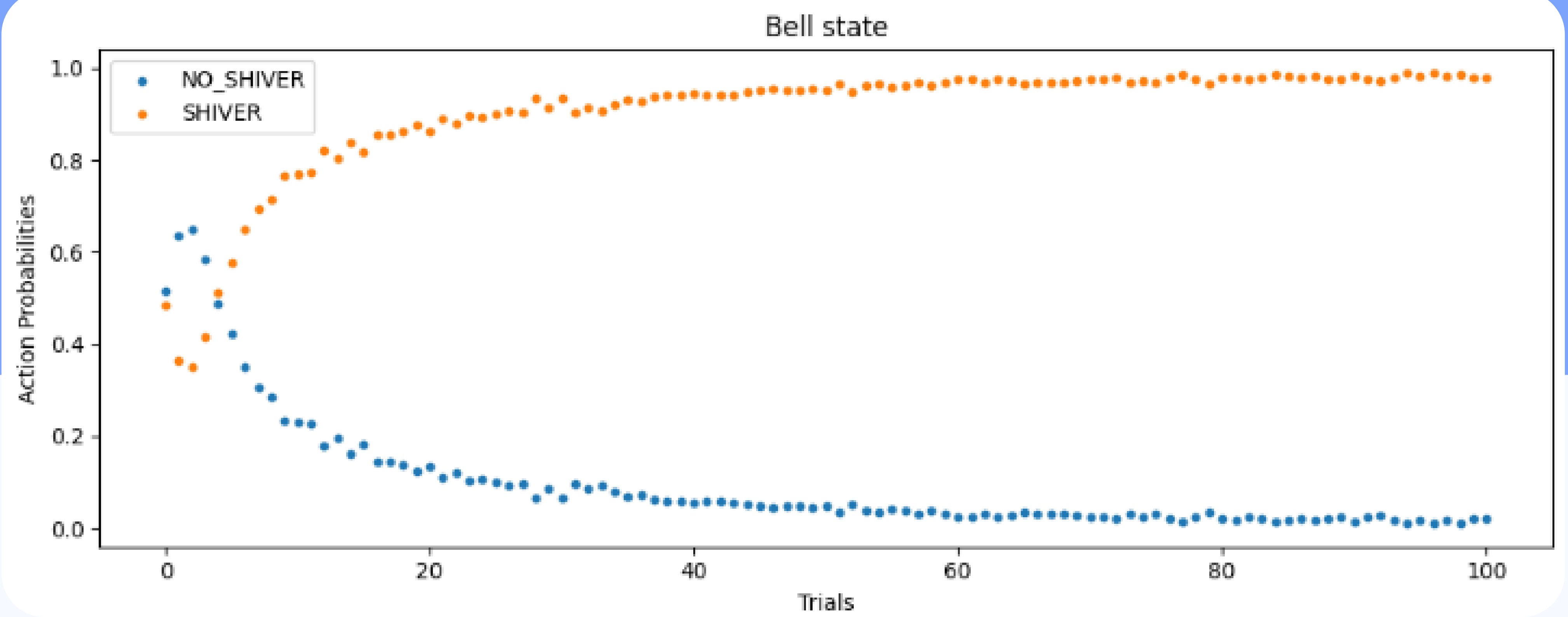


Our Plot

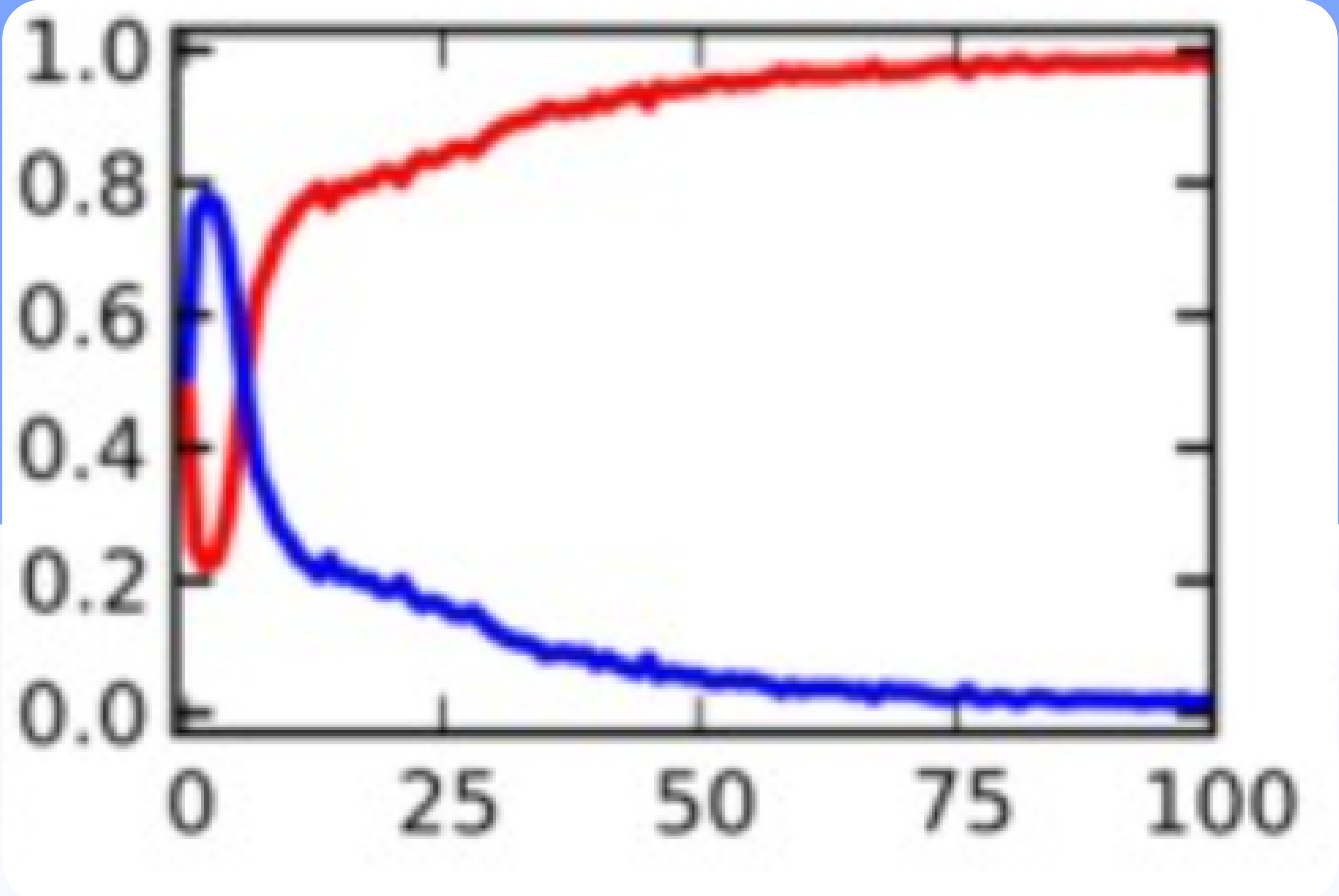


Paper's Plot

Results - Bell State

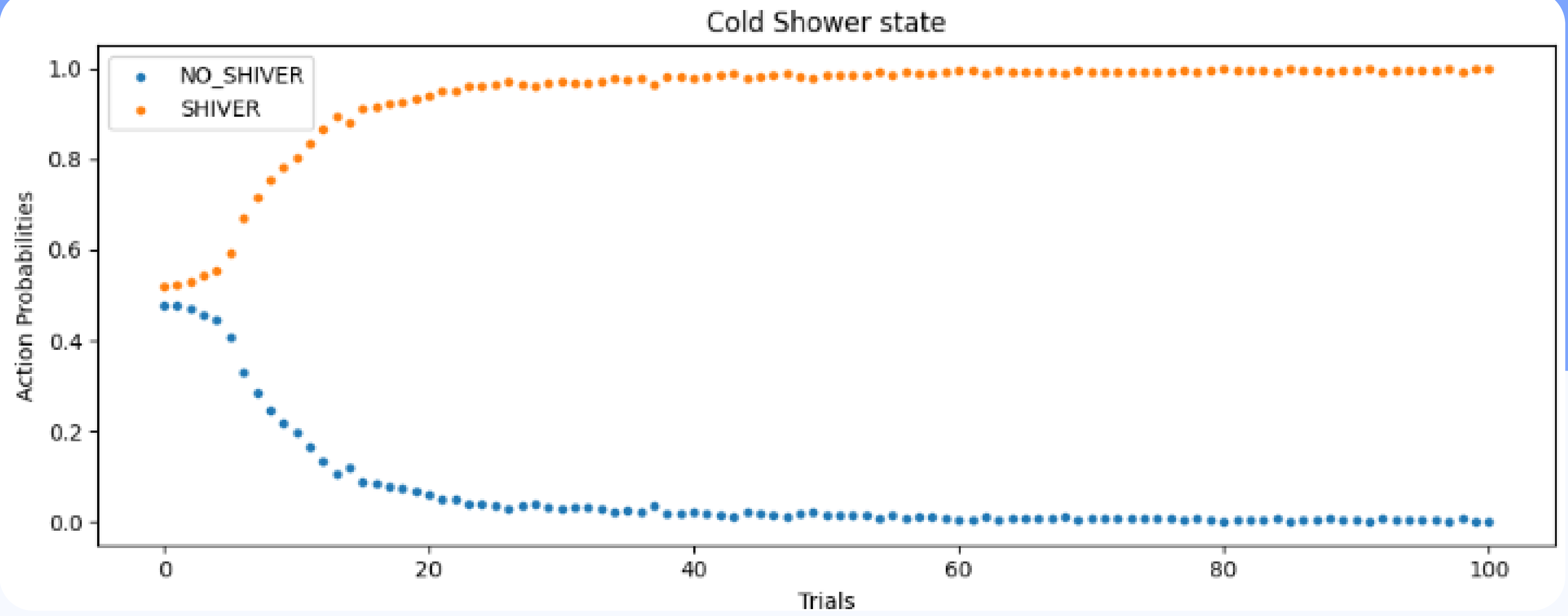


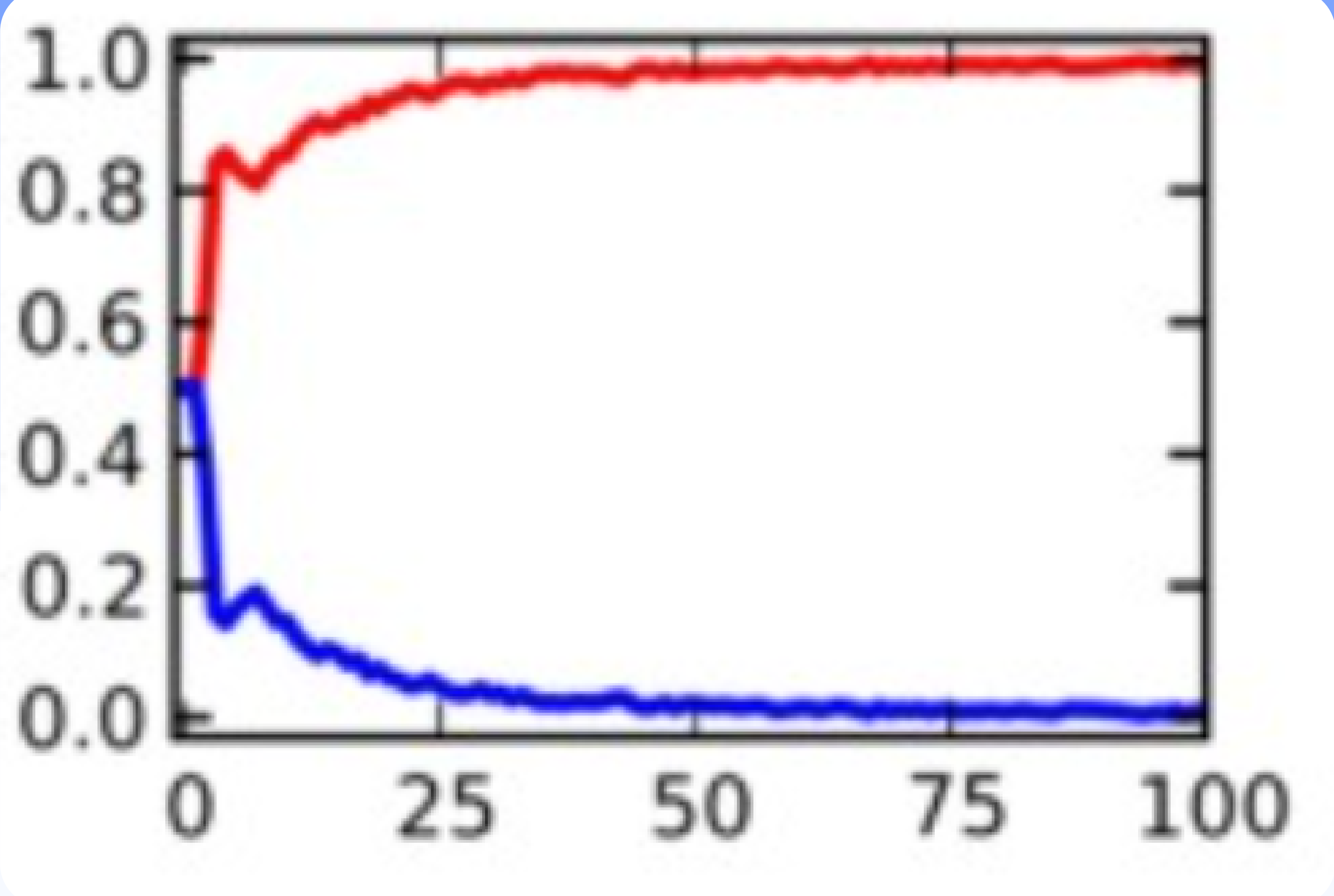
Our Plot



Paper's Plot

Results - Shower State





Our Plot

Paper's Plot

Thank you!

Any questions?